

数值分析第四次作业

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1.

(1) 讨论二次样条的存在唯一性 (严格证明);

(2) 讨论高次样条的可能性.

解答.

(1) 给定 $\{(x_j, y_j)\}_{0 \leq j \leq n}$, 设 $S(x) \in X_h^2 \cap C^1[a, b]$ 满足: $S(x_j) = y_j, \forall 0 \leq j \leq n$.

由于 $S'(x) \in X_h^1 \cap C[a, b]$, 故设

$$S'(x) |_{I_j} = M_{j-1} \frac{x_j - x}{h_j} + M_j \frac{x - x_{j-1}}{h_j}, \quad I_j := [x_{j-1}, x_j], \quad h_j := x_j - x_{j-1},$$

因此

$$S(x) |_{I_j} = -\frac{M_{j-1}}{2h_j}(x_j - x)^2 + \frac{M_j}{2h_j}(x - x_{j-1})^2 + C_j,$$

再分别代入 $x = x_j$ 和 $x = x_{j-1}$ 可得

$$y_j = \frac{1}{2}M_j h_j + C_j, \quad y_{j-1} = -\frac{1}{2}M_{j-1} h_j + C_j,$$

故

$$S(x) |_{I_j} = -\frac{M_{j-1}}{2h_j}(x_j - x)^2 + \frac{M_j}{2h_j}(x - x_{j-1})^2 + y_j - \frac{1}{2}M_j h_j,$$

其中 $M_0, M_1, \dots, M_n$ 满足:

$$M_j + M_{j-1} = \frac{2}{h_j}(y_j - y_{j-1}), \quad \forall 1 \leq j \leq n,$$

因此二次样条 $S(x)$ 显然存在且不唯一, 自由度为 1, 事实上, 此时 $S(x)$ 完全被 $M_0 = S'(x_0)$ 唯一确定.

(2) 设 $k \geq 2$, 给定 $\{(x_j, y_j)\}_{0 \leq j \leq n}$, 设 $S(x) \in X_h^k \cap C^{k-1}[a, b]$ 满足: $S(x_j) = y_j, \forall 0 \leq j \leq n$. 由于

$$\dim(X_h^k \cap C^{k-1}[a, b]) = n \cdot (k+1) - k(n-1) = n+k,$$

故给定 $n+1$ 个插值后自由度为

$$n+k-(n+1)=k-1 \geq 1,$$

因此 k 次样条 $S(x)$ 存在且不唯一, 自由度为 $k-1$. □

2. 对于三次样条函数 $S(x)$, 如果给定的插值条件是

$$S'(x_i) = y'_i, \quad 1 \leq i \leq n,$$

如何给出边界条件使得 $S(x)$ 被唯一确定.

解答.

再给定边界条件

$$S(x_0) = y_0, \quad S'(x_0) = y'_0, \quad S''(x_0) = y''_0,$$

下面证明此时 $S(x)$ 被唯一确定.

记二次样条 $T(x) = S'(x)$, 则 $T(x_j) = y'_j, \forall 0 \leq j \leq n$, 且 $T'(x_0) = y''_0$, 由 1(1) 知 $T(x)$ 被唯一确定, 因此三次样条

$$S(x) = \int_{x_0}^x T(s) ds + y_0$$

也被唯一确定. □

3. 验证在节点 $\{x_i\}_{0 \leq i \leq n}$ 上定义的规范化 B 样条基函数

$$B_{i,k}(x) := \frac{x_{i+k+1} - x_i}{k+1} M_{k+1}[x_i, \dots, x_{i+k+1}; x], \quad k \geq 0$$

满足下面的 Cox-de Boor 递推式

$$B_{i,0}(x) = \begin{cases} 1, & x \in [x_i, x_{i+1}), \\ 0, & \text{其他}, \end{cases}$$

$$B_{i,k}(x) = \frac{x - x_i}{x_{i+k} - x_i} B_{i,k-1}(x) + \frac{x_{i+k+1} - x}{x_{i+k+1} - x_{i+1}} B_{i+1,k-1}(x), \quad k \geq 1.$$

以及归一化性质

$$\sum_{i=J-k}^J B_{i,k}(x) = 1, \quad \forall x_J \leq x < x_{J+1}; \quad \sum_{i=-k}^{n-1} B_{i,k}(x) = 1, \quad \forall x_0 \leq x < x_n.$$

解答.

先证 Cox-de Boor 递推式. 由定义可得

$$\begin{aligned} B_{i,0}(x) &= (x_{i+1} - x_i)M_1[x_i, x_{i+1}; x] = M_1(x_{i+1}; x) - M_1(x_i; x) \\ &= (x_{i+1} - x)_+^0 - (x_i - x)_+^0 = \begin{cases} 1, & x \in [x_i, x_{i+1}), \\ 0, & \text{其他.} \end{cases} \end{aligned}$$

并且

$$\begin{aligned} & \frac{x - x_i}{x_{i+k} - x_i} B_{i,k-1}(x) + \frac{x_{i+k+1} - x}{x_{i+k+1} - x_{i+1}} B_{i+1,k-1}(x) \\ &= \frac{x - x_i}{x_{i+k} - x_i} \cdot \frac{x_{i+k} - x_i}{k} M_k[x_i, \dots, x_{i+k}; x] + \frac{x_{i+k+1} - x}{x_{i+k+1} - x_{i+1}} \cdot \frac{x_{i+k+1} - x_{i+1}}{k} M_k[x_{i+1}, \dots, x_{i+k+1}; x] \\ &= \frac{x - x_i}{k} M_k[x_i, \dots, x_{i+k}; x] + \frac{x_{i+k+1} - x}{k} M_k[x_{i+1}, \dots, x_{i+k+1}; x], \end{aligned}$$

以及

$$B_{i,k}(x) = \frac{x_{i+k+1} - x_i}{k+1} M_{k+1}[x_i, \dots, x_{i+k+1}; x]$$

故只需证:

$$\frac{x_{i+k+1} - x_i}{k+1} M_{k+1}[x_i, \dots, x_{i+k+1}; x] = \frac{x - x_i}{k} M_k[x_i, \dots, x_{i+k}; x] + \frac{x_{i+k+1} - x}{k} M_k[x_{i+1}, \dots, x_{i+k+1}; x].$$

从而也只需证:

$$(x_{i+k+1} - x_i) \sum_{j=i}^{i+k+1} \frac{(x_j - x)_+^k}{\omega'_{k+2}(x_j)} = (x - x_i) \sum_{j=i}^{i+k} \frac{(x_j - x)_+^{k-1}}{\omega'_{k+2}(x_j)/(x_j - x_{i+k+1})} + (x_{i+k+1} - x) \sum_{j=i+1}^{i+k+1} \frac{(x_j - x)_+^{k-1}}{\omega'_{k+2}(x_j)/(x_j - x_i)},$$

因此只需证: $\forall i+1 \leq j \leq i+k$, 都有

$$\begin{aligned} (x_{i+k+1} - x_i)(x_j - x)_+^k &= (x - x_i)(x_j - x_{i+k+1})(x_j - x)_+^{k-1} + (x_{i+k+1} - x)(x_j - x_i)(x_j - x)_+^{k-1}; \\ (x_{i+k+1} - x_i)(x_i - x)_+^k &= (x - x_i)(x_i - x_{i+k+1})(x_i - x)_+^{k-1}; \\ (x_{i+k+1} - x_i)(x_{i+k+1} - x)_+^k &= (x_{i+k+1} - x)(x_{i+k+1} - x_i)(x_{i+k+1} - x)_+^{k-1}. \end{aligned}$$

对于第一个式子, 若 $x \geq x_j$, 则两边显然都等于 0; 若 $x < x_j$, 则等价于

$$(x_{i+k+1} - x_i)(x_j - x) = (x - x_i)(x_j - x_{i+k+1}) + (x_{i+k+1} - x)(x_j - x_i),$$

也显然成立. 对于第二、三个式子同理可证.

因此 Cox-de Boor 递推式得证, 由此归纳易证 $\text{supp } B_{i,k}(x) = [x_i, x_{i+k+1})$.

再证归一化性质. 先证第一个式子, 对 $k \geq 0$ 归纳, 当 $k = 0$ 时显然, 假设命题对 $k-1$ 成立, 其中 $k \geq 1$,

对 k : 由 Cox-de Boor 递推式, $\text{supp } B_{i,k}(x) = [x_i, x_{i+k+1})$ 以及归纳假设可得

$$\begin{aligned}\sum_{i=J-k}^J B_{i,k}(x) &= \sum_{i=J-k}^J \left(\frac{x - x_i}{x_{i+k} - x_i} B_{i,k-1}(x) + \frac{x_{i+k+1} - x}{x_{i+k+1} - x_{i+1}} B_{i+1,k-1}(x) \right) \\ &= \sum_{i=J-k+1}^J \frac{x - x_i}{x_{i+k} - x_i} B_{i,k-1}(x) + \sum_{i=J-k+1}^J \frac{x_{i+k} - x}{x_{i+k} - x_i} B_{i,k-1}(x) = \sum_{i=J-k+1}^J B_{i,k-1}(x) = 1,\end{aligned}$$

从而命题对 k 也成立, 由数学归纳法知第一个式子成立. 再结合 $\text{supp } B_{i,k}(x) = [x_i, x_{i+k+1})$ 知第二个式子也成立. $\square$